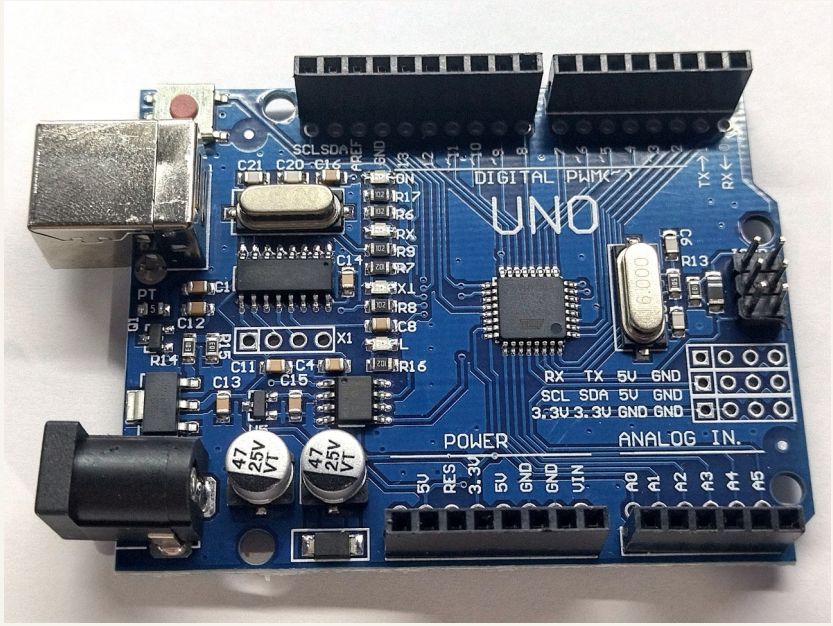
# Daily Review

| | | |
|---|---|---|
| 1 | A transistor in a circuit acts as a… | Switch |
| 2 | A relay lets a small current switch… | A separate, bigger circuit |
| 3 | A diode across a relay coil protects the… | Transistor |
| 4 | It gets dark. An LDR's resistance… | Rises |
| 5 | Three parts of a system diagram? | Input, process, output |

# Microcontrollers

One chip runs a washing machine, a bus door and a packing line.




National 5 Engineering Science

# How does a washing machine know when to spin?




A washing machine

PART 1

# What a Microcontroller Is

One chip, one program, a whole machine.

# By the end of this part…



1. I can describe what a microcontroller is
2. I can give commercial and industrial examples
3. I can give advantages and disadvantages against hard-wired control

# What Is a Microcontroller?

## Microcontroller

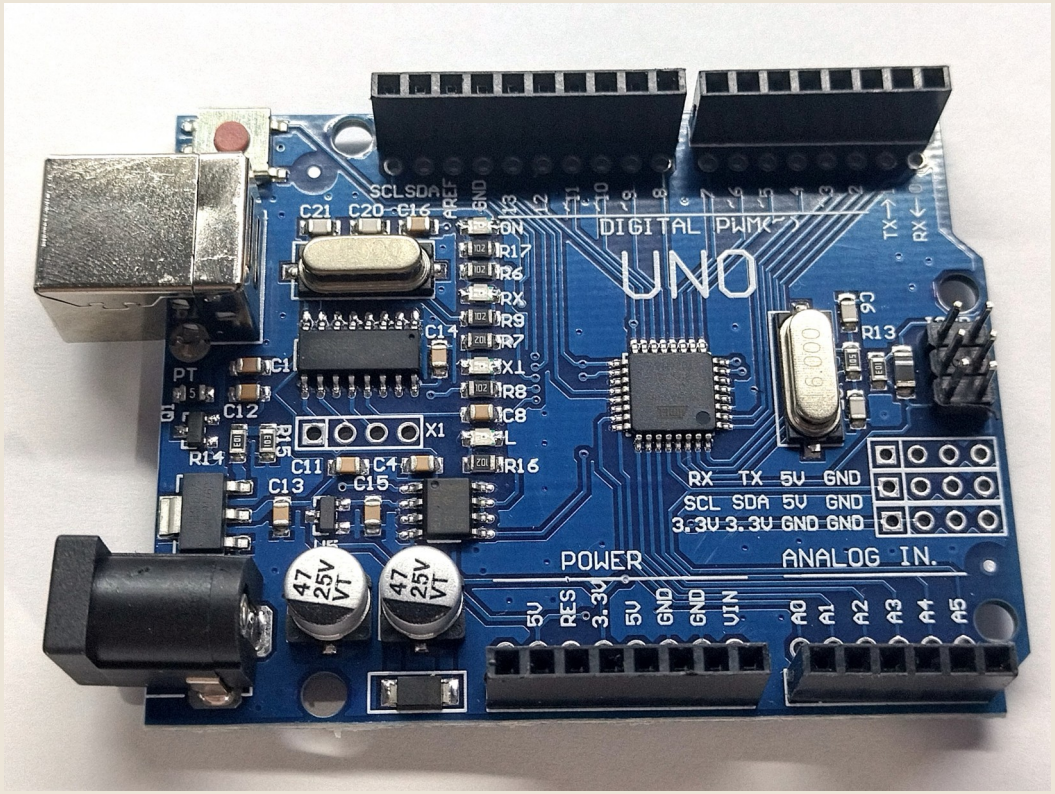
a small, programmable chip that controls a system

## It runs

a program stored in its memory

## It reads

inputs, and switches outputs



# Input, Process, Output

**Input**

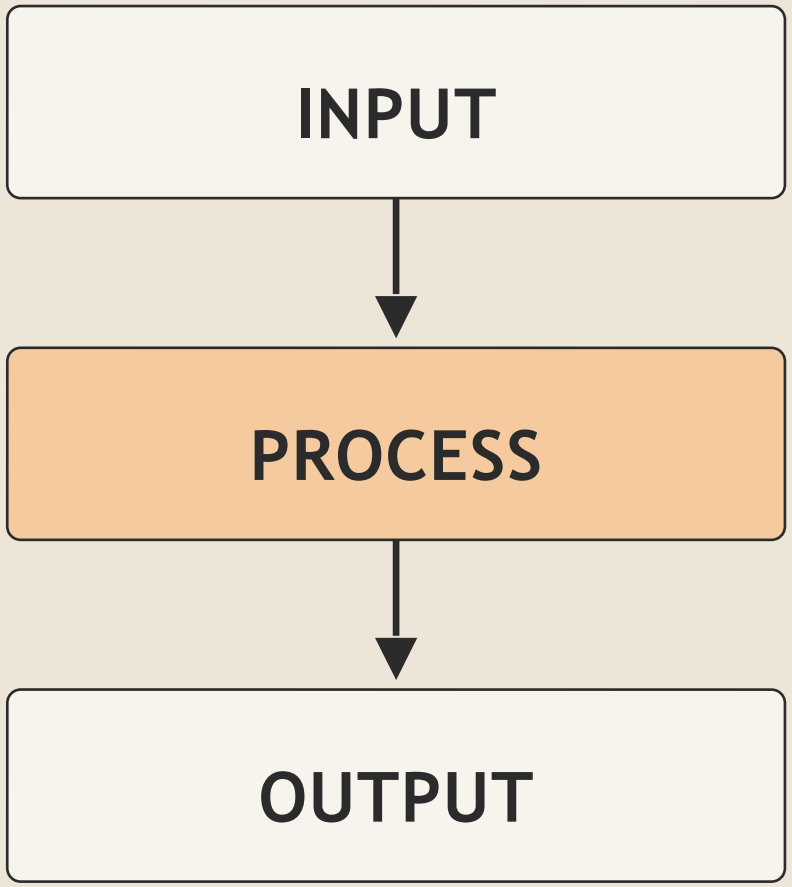
a switch or a sensor

**Process**

the microcontroller and its program

**Output**

a lamp, buzzer or motor



# Where You Meet Them




## Commercial

ticket machine




## Industrial

packing line



## Transport

bus doors




## Home

smart lighting

# Which Sector?

| | | |
|---|---|---|
| 1 | An alarm panel in a shop | Commercial |
| 2 | A lift in an office block | Industrial |
| 3 | Anti-lock brakes on a car | Transport |
| 4 | Central heating control | Home |
| 5 | A packaging machine | Industrial |

# Hard-Wired Control

**Hard-wired**

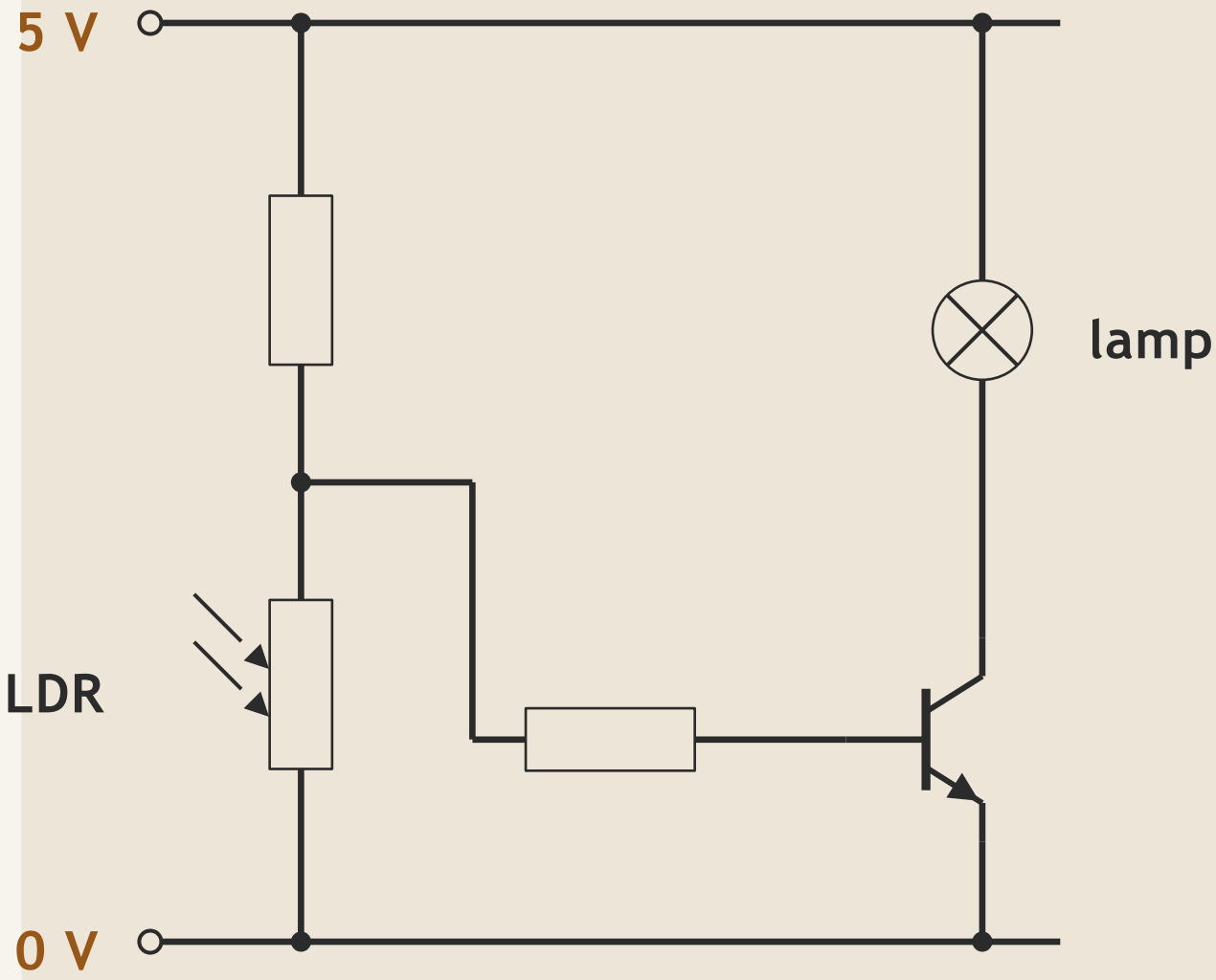
fixed components and wires

**The job is**

in the wiring

**Change the job**

new parts, new wiring



# Programmable Control

**Programmable**

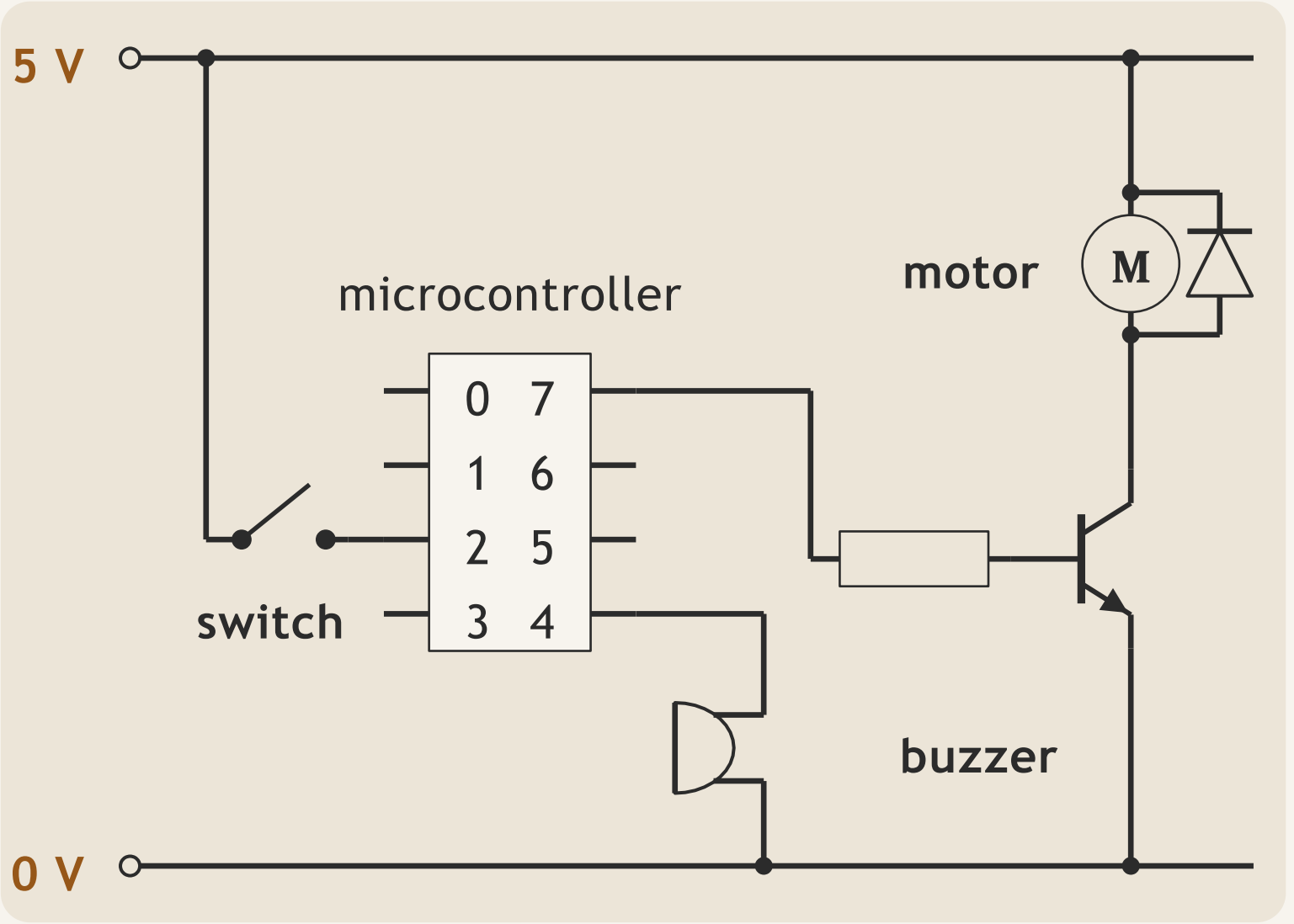
a microcontroller and a program

**The job is**

in the program

**Change the job**

change the program



# Advantages

## Reprogrammed

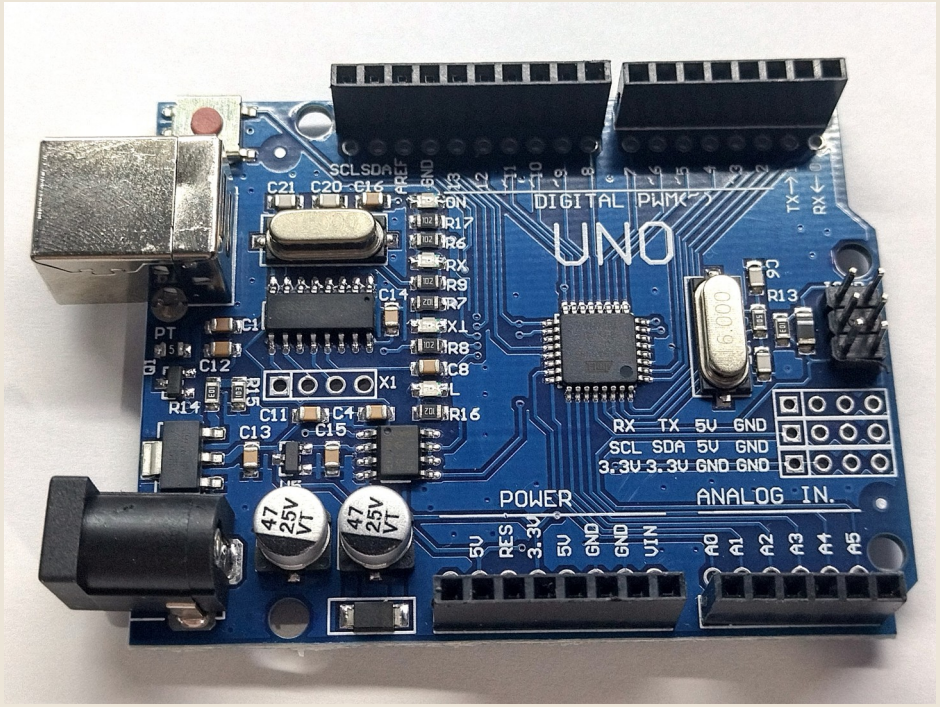
change the program, not the wiring

## Re-used

one board, many products

## Fewer components

so more reliable, faster to assemble



# Disadvantages

## Needs a programmer

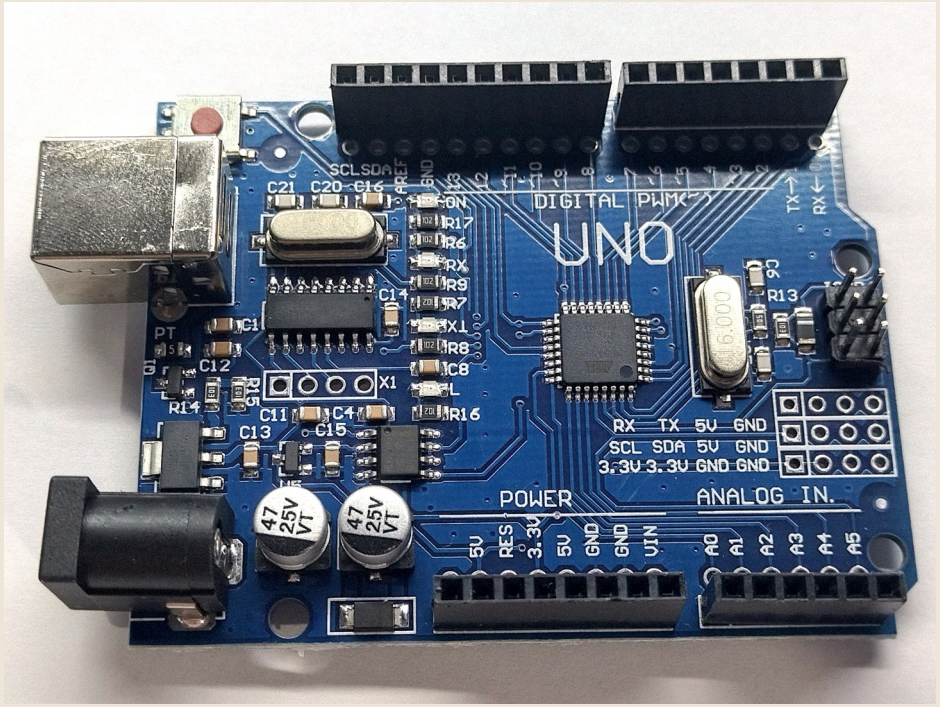
someone must write and test the program

## Needs a power supply

the chip needs one to run

## Harder to follow

you cannot see the job by looking at the parts



# Watch Out: “Easier to Change”

**THE MISTAKE**

- It is easier to change.
- It is cheaper.
- It is programmable.

**THE CORRECT ANSWER**

- It can be reprogrammed, with no rewiring.
- It needs fewer components, so it costs less.
- One board can be re-used for a new job.

# Model Answer: Describe Two Advantages (2 marks)

Describe two advantages of a microcontroller over a hard-wired circuit for a bus's doors.

**1 (1 mark)**

The door times can be reprogrammed with no rewiring.

**2 (1 mark)**

Fewer components, so the doors are more reliable.



# Model Answer: Explain Why (2 marks)

Explain why a microcontroller is used in a ticket machine.

## Cause (1 mark)

The program can be updated

## Effect (1 mark)

so new fares are added with no change to the circuit.



# Your Turn: Same Form




1. Smart lighting: describe one advantage.

Reprogrammed — **so no rewiring**

2. A packing line: explain why (cause + effect).

Fewer parts — **so more reliable**

3. A pupil writes 'it is cheaper'. Marks?

0 on its own: say why it costs less

# Spot the Mistake

Describe two advantages of a microcontroller for a car-park barrier.

**A pupil wrote**

1 It is cheaper.  2 It is programmable.

**Your job**

How many marks? Why? Rewrite it.



# Booklet page 3, TASK 1

For each product, say why a microcontroller suits it.
Use the advantage words.

5 min

# TASK 1: Answers

| | | |
|---|---|---|
| 1 | Microwave oven | Timed sequence; reads the door |
| 2 | Traffic lights | Timings changed by program, not rewiring |
| 3 | Robot arm in a factory | Runs motors in order; reprogrammed |
| 4 | Smart heating thermostat | Reads a sensor; settings set by program |

# Past Papers: Advantages

## Past papers

| Year | Question |
|---|---|
| 2021 | Q7(b) |
| 2022 | Q14(c) |
| 2025 | Q8(c) |
| 2026 | Q13(e) |

By topic

# Checkpoint




1. What is a microcontroller?

2. Give one disadvantage.

3. Why use one board for 3 alarm models?

A chip that runs a stored program

Someone must write the program

Re-usable — **so just a new program**

# Plenary: Part 1




1. A microcontroller runs a stored…

Program

2. Give an industrial example.

Packing line, lift, robot arm

3. Two advantages, MI words?

Reprogrammed; fewer parts, so reliable

# Plenary: Part 1



4 One disadvantage?

Needs a programmer (or power supply)

5 What scores 0 on its own?

Easier to change, cheaper, fewer parts

# Look back — by the end of this part…



1. I can describe what a microcontroller is
2. I can give commercial and industrial examples
3. I can give advantages and disadvantages against hard-wired control

# Daily Review



| | | |
|---|---|---|
| 1 | One advantage of a microcontroller? | It can be reprogrammed |
| 2 | 'Easier to change' alone scores… | 0 marks |
| 3 | A transistor switches on at a base voltage of… | About 0.7 V |
| 4 | The base resistor limits the… | Base current |
| 5 | A packing line: commercial or industrial? | Industrial |

PART 2

# Wiring to the Pins

The lamp is too dim. What do you add?

# By the end of this part…



1. I can wire an input switch to a pin
2. I can wire an output to a pin
3. I can wire a transistor driver to a pin
4. I can complete a wiring diagram from an I/O table

# The Exam's Microcontroller

**8 pins**

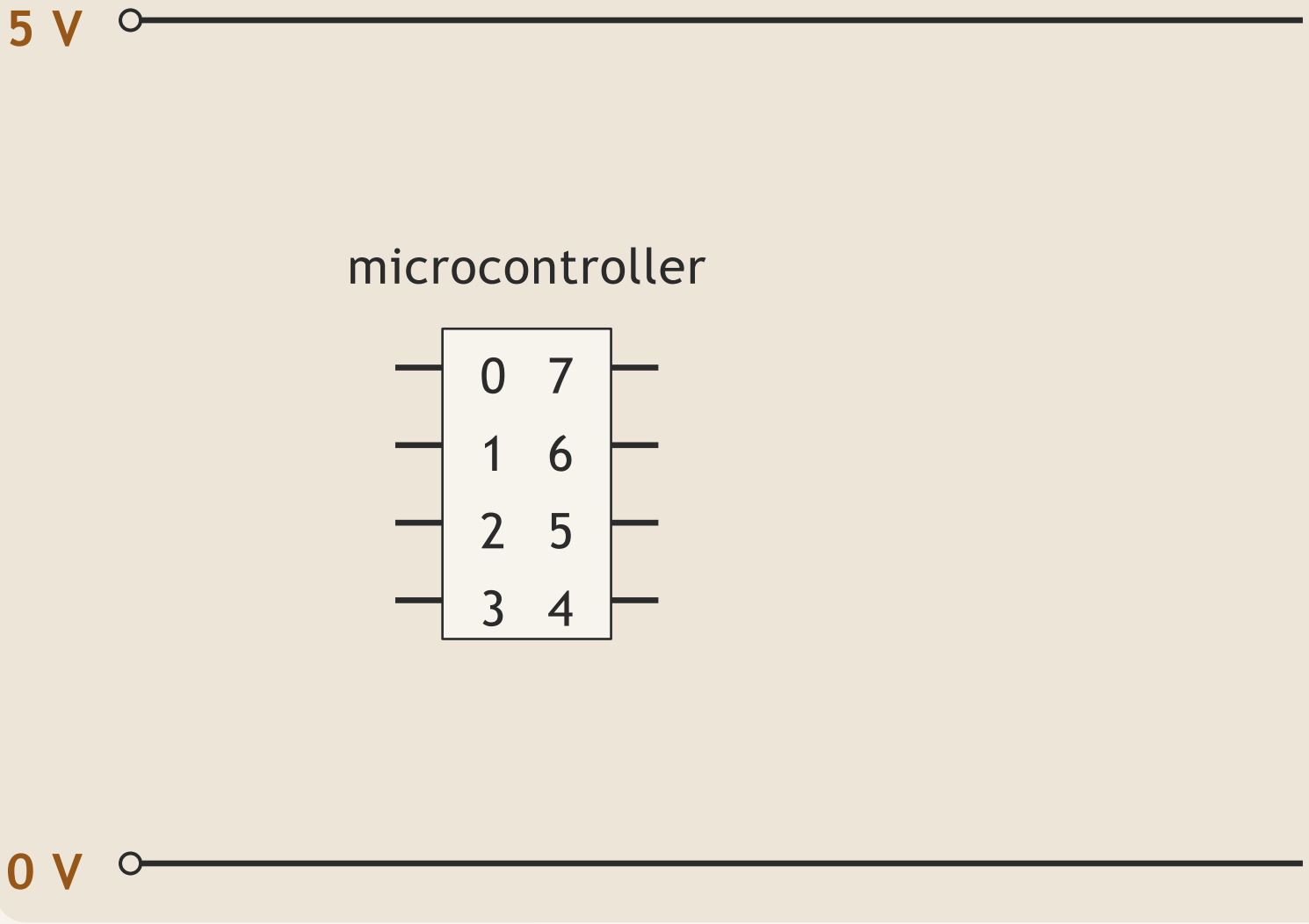
each an input or an output

**The I/O table**

says which

**In our room**

in: 2, 3  out: 4 to 7



# Wiring an Input

**Switch**

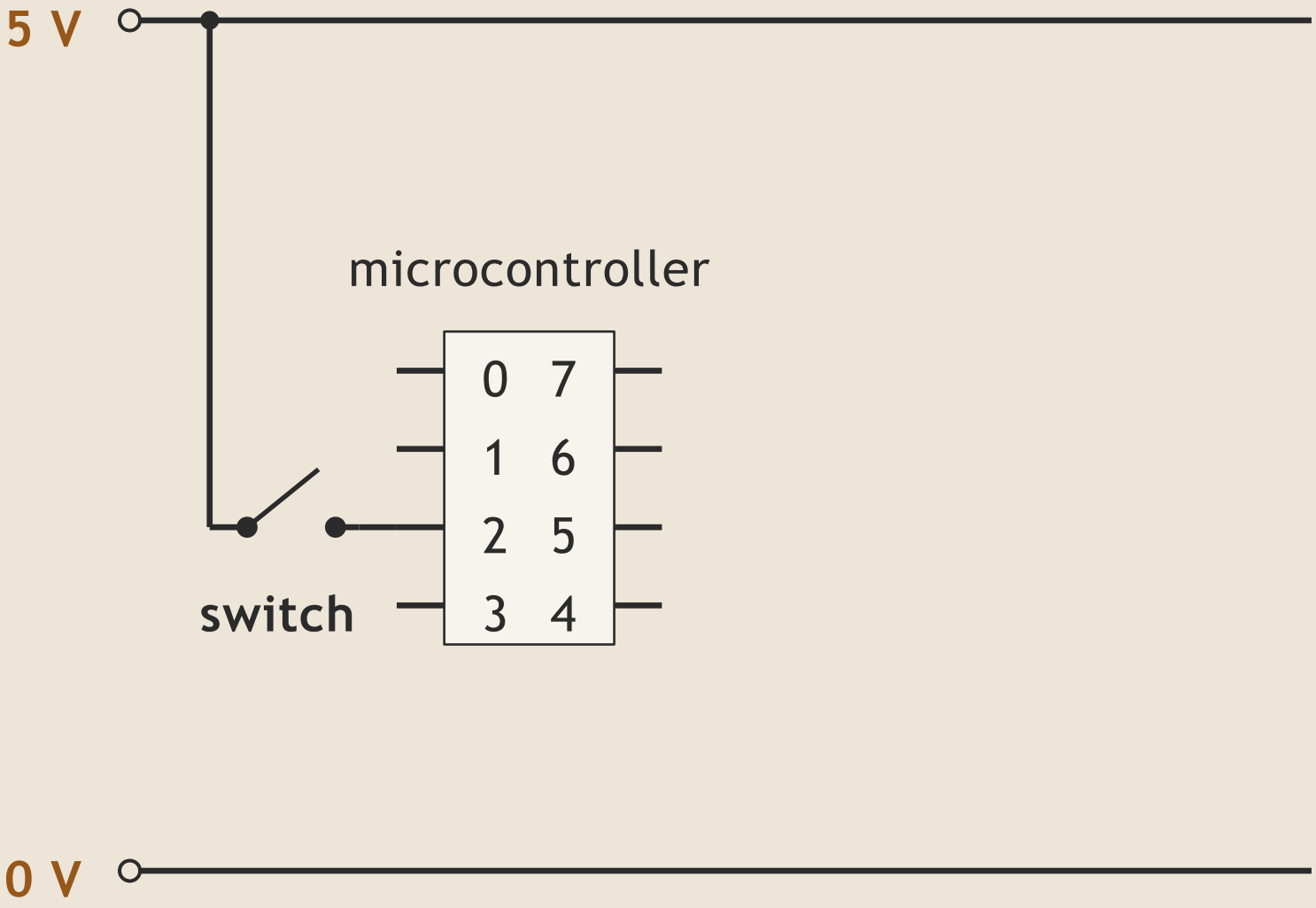
from 5 V to the pin

**Pressed**

the pin gets 5 V

**The program**

reads the input as on



# Wiring an Output

**Output**

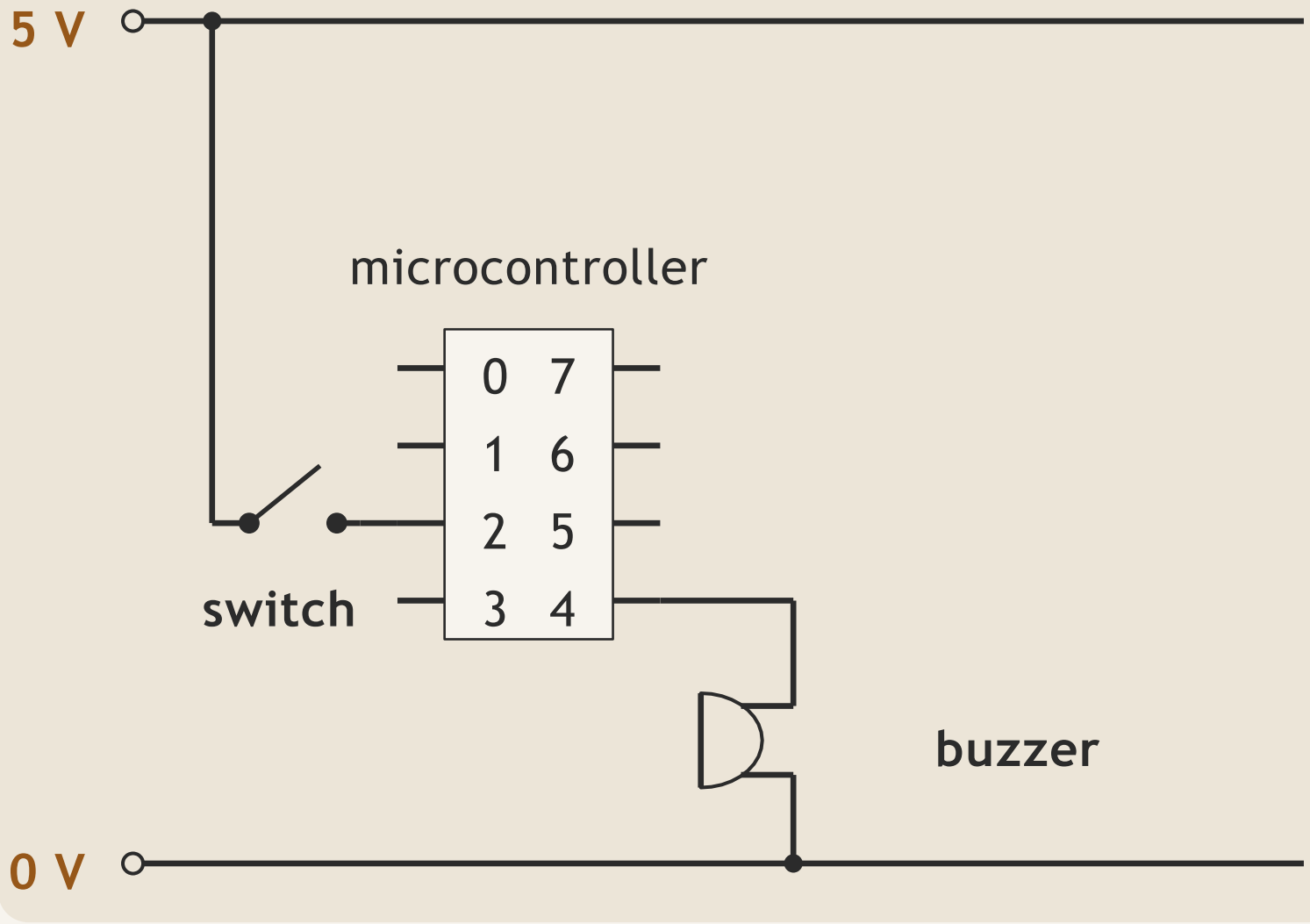
from the pin to 0 V

**Switched on**

5 V on the pin

**Current flows**

the buzzer sounds



# A Motor Needs a Transistor

**A pin gives**

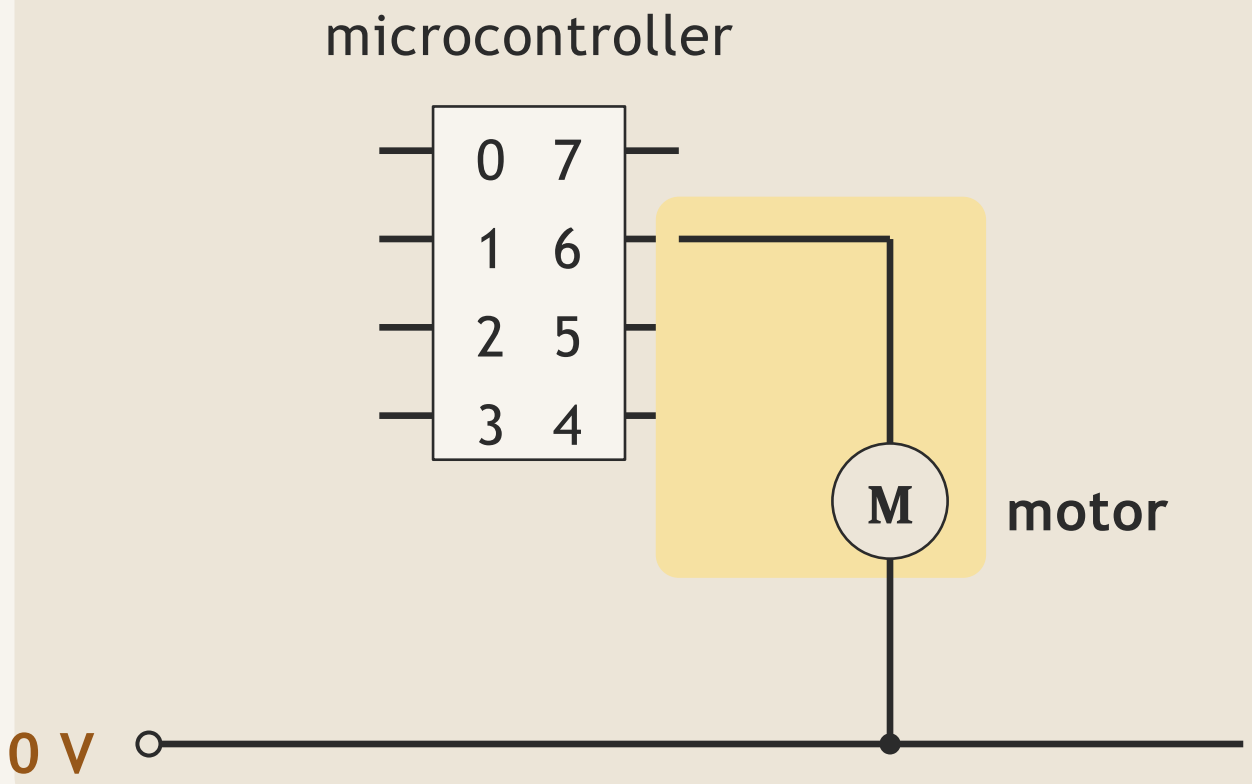
a small current, about 20 mA

**A motor needs**

much more

**Add a transistor**

“driver” alone scores 0



# The Transistor Driver

**Base lead**

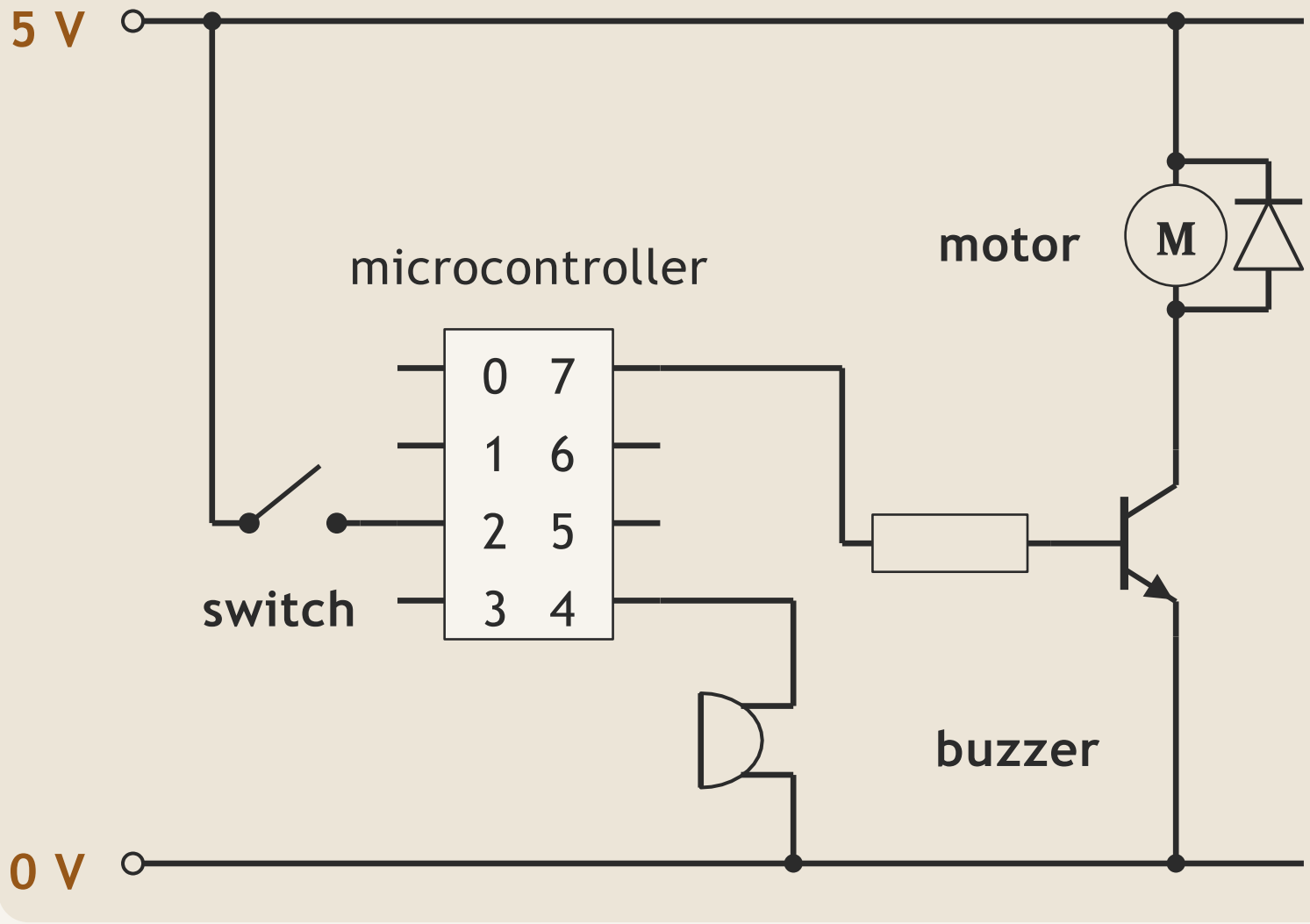
pin, resistor, base

**Collector**

motor, diode across it

**Emitter**

to 0 V



# Model Answer: Complete the Wiring (3 marks)

**Input, pin 2**

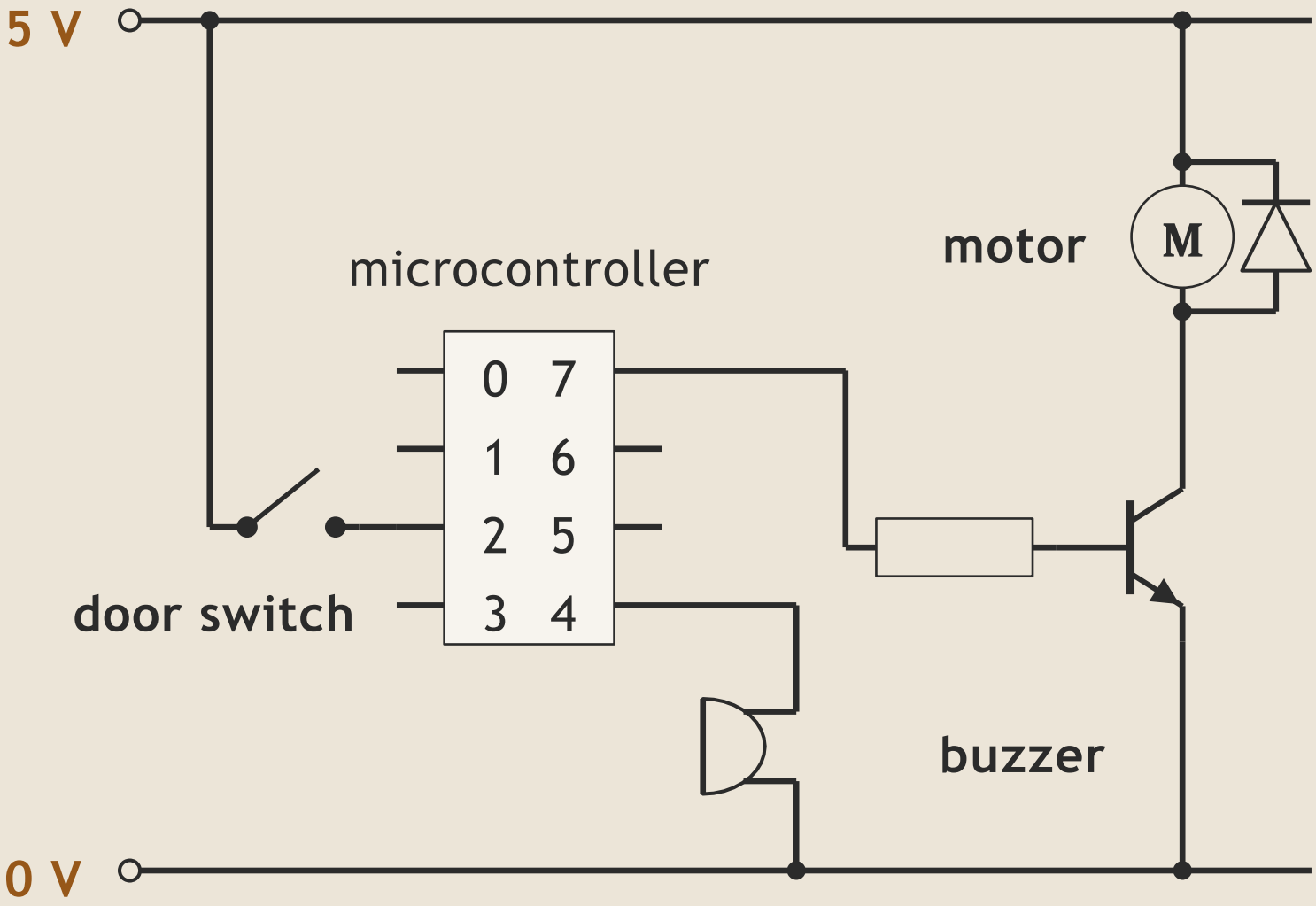
door switch

**Output, pin 4**

buzzer

**Output, pin 7**

motor, via a transistor



# The Same Circuit in the Simulator

## Same I/O table

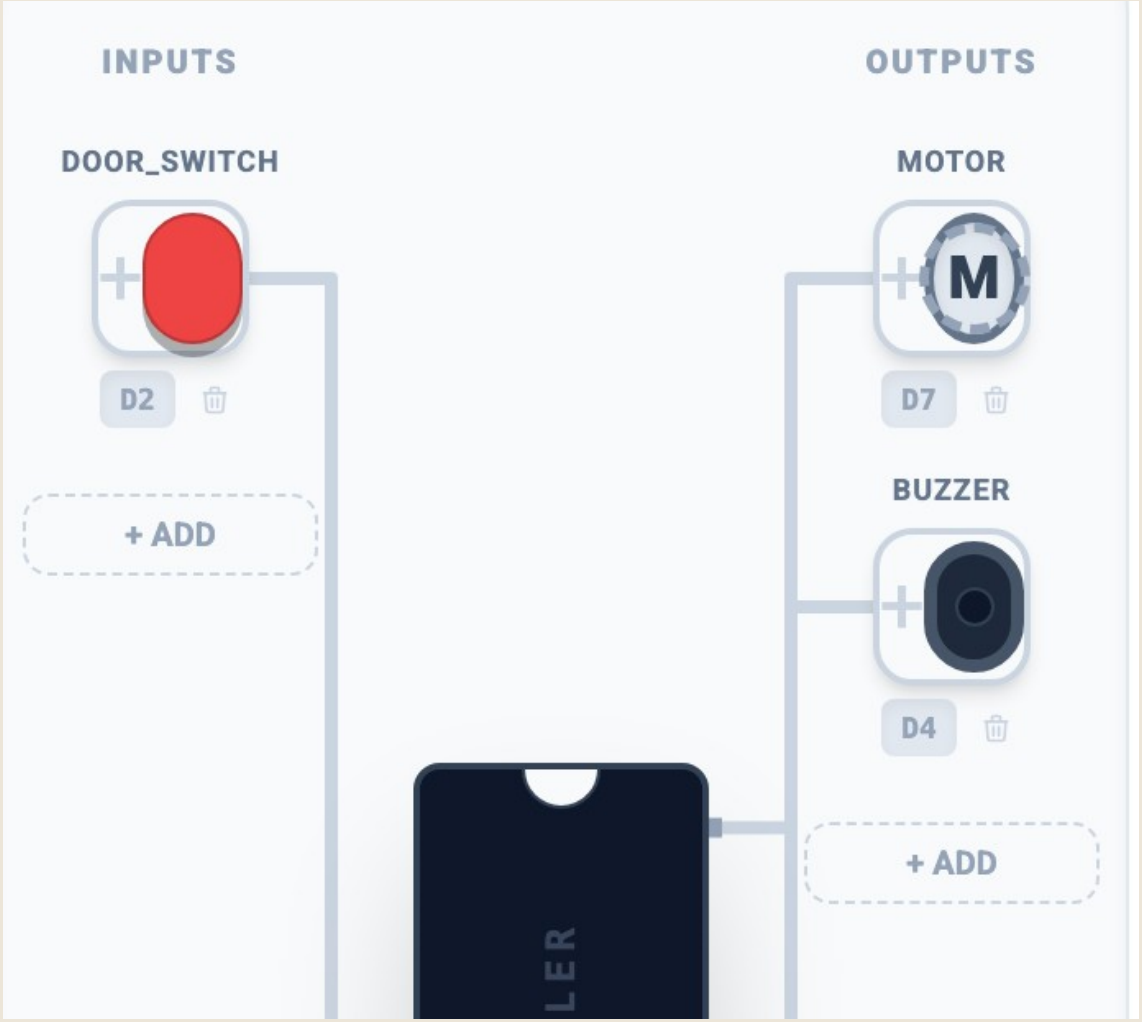
D2 in; D4 and D7 out

## Each part

on its own pin

## The sim hides

the transistor and the rails



# Your Turn: Complete the Wiring

**Input, pin 3**

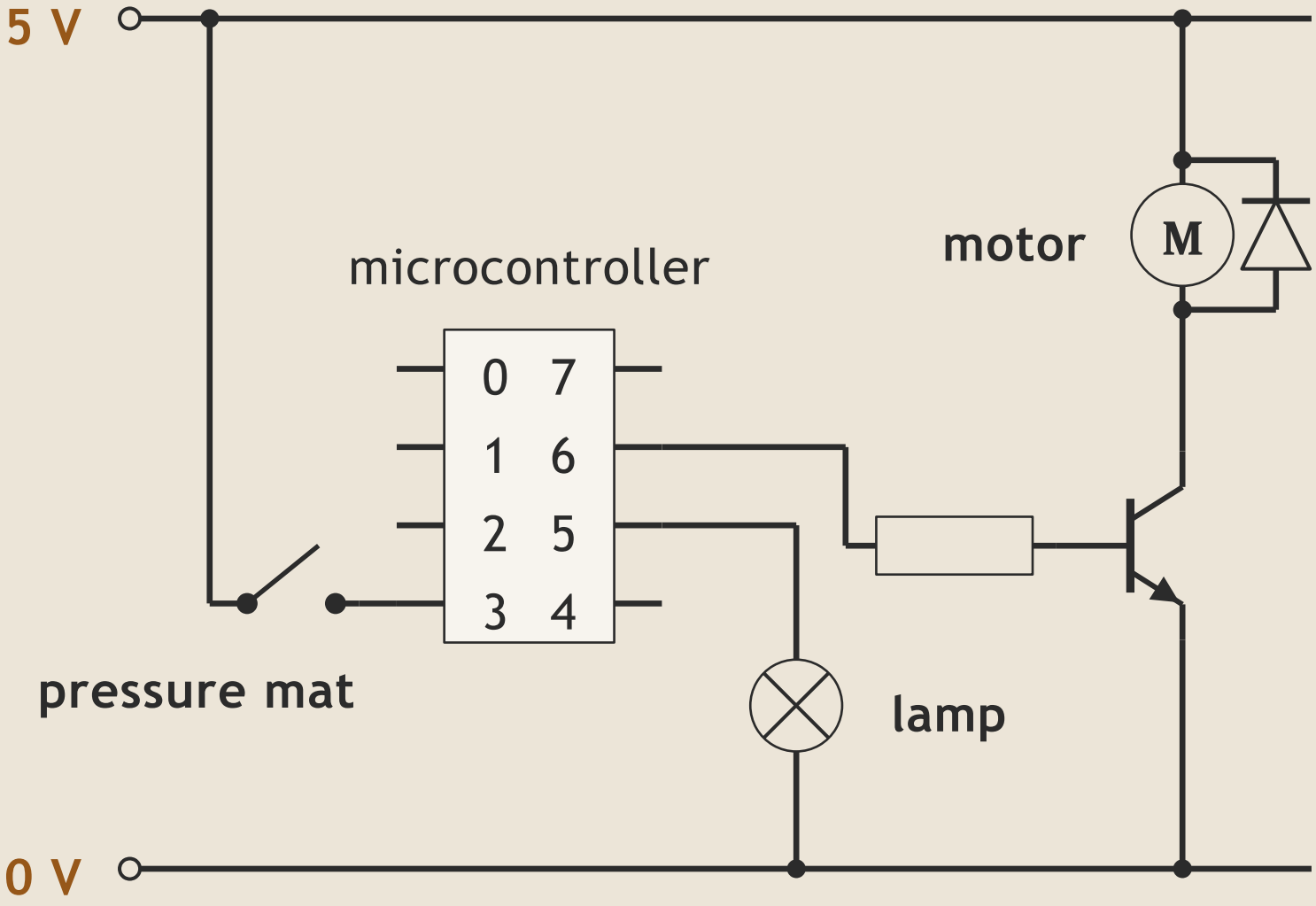
pressure mat

**Output, pin 5**

lamp

**Output, pin 6**

motor, via a transistor



# Spot the Mistake

**I/O table**

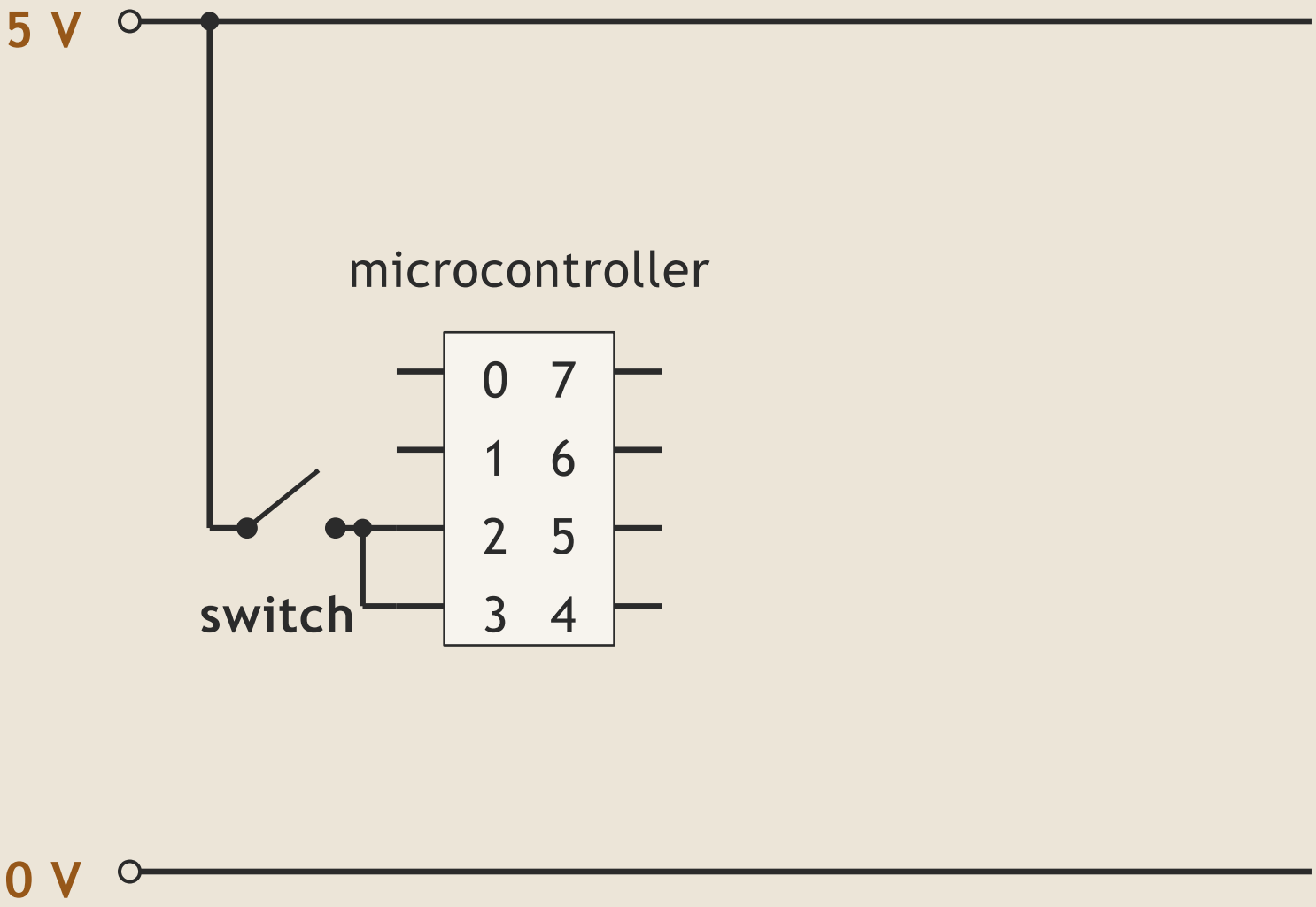
switch on pin 2

**A pupil drew**

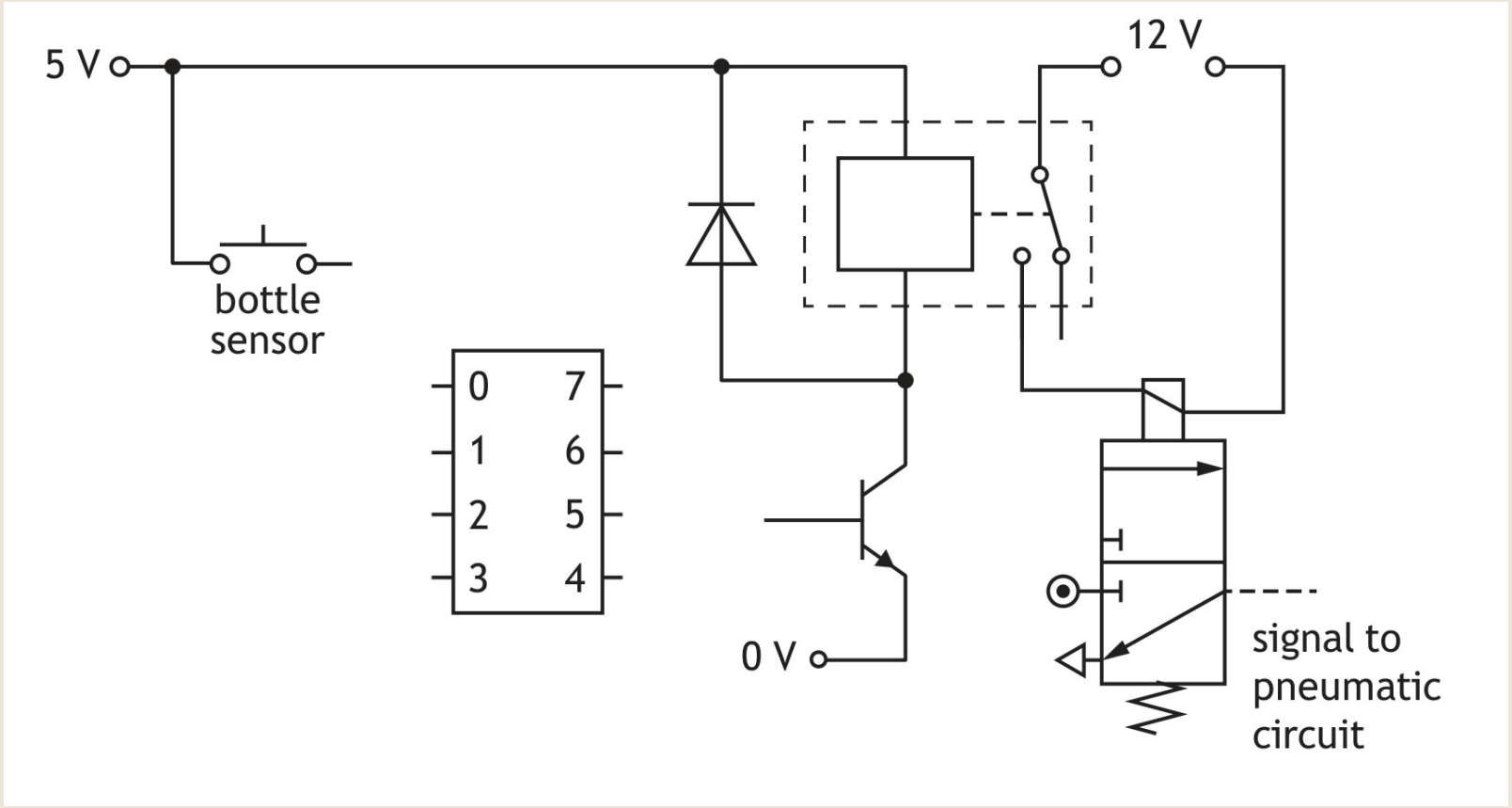
this

**Your job**

find the mistake;
what does it cost?



# Past Papers: Wiring




2023 Q9(d) (2 marks)

# Past Papers: Answers

1. 2023 Q9(d)

Sensor to pin 2 (1); pin 7 to base (1)

2. 2023 Q9(d), also

No extra pins wired

# Checkpoint




1. Where does an input switch connect?

   From 5 V to the pin

2. Where does a buzzer connect?

   From the pin to 0 V

3. Why can't a motor go straight on a pin?

   Too little current — **so a transistor**

# Plenary: Part 2

| | | |
|---|---|---|
| 1 | An input switch goes from … to the pin. | 5 V |
| 2 | A small output goes from the pin to… | 0 V |
| 3 | A motor needs what between it and the pin? | A transistor |
| 4 | What goes across the motor? | A protection diode |
| 5 | A wire to a pin not in the I/O table… | Loses the mark |

# Extension

EXTRA PRACTICE




1. Why is a diode put across the motor?

   Protects the transistor from the spike

2. Why the resistor in the base lead?

   It limits the base current

3. A 12 V lamp on pin 5: straight on, or not?

   Through a transistor, with a 12 V supply

# Look back — by the end of this part…



1. I can wire an input switch to a pin
2. I can wire an output to a pin
3. I can wire a transistor driver to a pin
4. I can complete a wiring diagram from an I/O table

LOOKING AHEAD

# Lesson 2: Flowcharts

The wiring is done. How do we tell the chip what to do?

# Course Specification

## Digital electronic control systems

- examples of using microcontrollers in commercial and industrial applications
- advantages and disadvantages of microcontroller-based control systems, compared to a hard-wired electronic equivalent

Wiring to pins: asked in the 2023 and 2025 papers.

# Credits (1 of 4)

**The big question (A washing machine):** Photo: Fanti Salms, CC BY-SA 3.0, via Wikimedia Commons

**Title slide (Arduino Uno board):** Photo: JrawX, CC0, via Wikimedia Commons

**What Is a Microcontroller? (Arduino Uno board):** Photo: JrawX, CC0, via Wikimedia Commons

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**Where You Meet Them (packing line):** Photo: SimónK, CC BY-SA 4.0, via Wikimedia Commons

**Where You Meet Them (bus doors):** Photo: Leif Jørgensen, CC BY-SA 4.0, via Wikimedia Commons

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**Model Answer: Complete the Wiring (2023 Q9(d) MI):** © Qualifications Scotland (SQA). N5 Engineering Science 2023 Q9(d) and 2025 Q9(b)(c), question papers and marking instructions.

**The Same Circuit in the Simulator:** NoSyntaxSim V2 screenshot (hardware panel). Simulator by Peter Strain, NoStrainSims (sites.google.com/view/nostrainsim), free and open to all.

**Spot the Mistake (2023 Q9(d) MI):** © Qualifications Scotland (SQA). N5 Engineering Science 2023 Q9(d) and 2025 Q9(b)(c), question papers and marking instructions.

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**Past Papers: Wiring:** © Qualifications Scotland (SQA). N5 Engineering Science 2023 Q9(d), question paper and marking instructions.

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R Stewart - October 2026